

AULA BÔNUS 12

Prof. Sérgio Altenfelder

- TRANSFORMAÇÃO DE DÍZIMA PERÍODICA EM FRAÇÃO

Dso

DIREITO SIMPLES E OBJETIVO

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$0,11111\dots = \frac{1}{9}$ ✓
 $0,\overline{1}$

$0,22222\dots = \frac{2}{9}$ ✓
 $0,\overline{2}$

$0,44444\dots = \frac{4}{9}$ ✓
 $0,\overline{4}$

$0,77777\dots = \frac{7}{9}$ ✓
 $0,\overline{7}$

$0,121212\dots = \frac{12 \div 3}{99 \div 3} = \frac{4}{33}$ ✓
 $0,\overline{12}$

$0,232323\dots = \frac{23}{99}$ ✓
 $0,\overline{23}$

$0,373737\dots = \frac{37}{99}$ ✓
 $0,\overline{37}$

$0,137137137\dots = \frac{137}{999}$ ✓
 $0,\overline{137}$

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$$\begin{array}{l} 0,011111... = \frac{1}{90} \checkmark \\ \text{0,01}\overline{1} \end{array}$$
$$\begin{array}{l} 0,002222... = \frac{2 \div 2}{900 \div 2} = \frac{1}{450} \\ \text{0,00}\overline{2} \end{array}$$
$$\begin{array}{l} 0,00044444... = \frac{4 \div 4}{9000 \div 4} = \frac{1}{2250} \\ \text{0,000}\overline{4} \end{array}$$
$$\begin{array}{l} 0,0121212... = \frac{12 \div 6}{990 \div 6} = \frac{2}{165} \\ \text{0,01}\overline{2} \end{array}$$

$$\begin{array}{l} 0,0232323... = \frac{23}{990} \checkmark \\ \text{0,02}\overline{3} \end{array}$$
$$\begin{array}{l} 13,22222... = 13 + 0,222... \\ \text{13,2}\overline{2} \end{array}$$
$$\begin{array}{l} 13 + \frac{2}{9} \\ \hline 119 \\ 9 \end{array}$$
$$\frac{119}{9} \checkmark$$

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$$\begin{array}{l} 1,22222... = 1 + 0,222... = 1 + \frac{2}{9} = \frac{11}{9} \\ \text{1,2}\overline{2} \end{array}$$
$$\begin{array}{l} \bullet \text{1,34444...} = \underline{1,3} + \text{0,0444...} = \frac{13}{10} + \frac{4}{90} = \frac{121}{90} \\ \text{1,3}\overline{4} \end{array}$$
$$\begin{array}{l} 2,131313... = 2 + 0,131313... \\ \text{2,13}\overline{13} \end{array}$$
$$\begin{array}{l} 2,213333... = \underline{2,21} + \text{0,00333...} \\ \text{2,21}\overline{3} \end{array}$$

$$\begin{array}{l} 2 + \frac{13}{99} = \frac{211}{99} \\ \frac{221}{100} + \frac{3}{900} = \frac{1992 \div 6}{900 \div 6} = \frac{332}{150} \div 2 = \frac{166}{75} \end{array}$$

$$\begin{array}{r} 1989 \\ 3 \\ \hline 1992 \\ 19 \quad 12 \\ \hline 221 \\ 9 \\ \hline 1989 \end{array}$$
$$\begin{array}{r} 99 \\ \times 2 \\ \hline 198 \\ 13 \\ \hline 211 \end{array}$$

EXERCÍCIO DE FIXAÇÃO

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1. Transforme as dízimas periódicas em fração simplificando se possível:

a) $0,22222... = \frac{2}{9}$

b) $0,55555... = \frac{5}{9}$

c) $0,121212... = \frac{12 \div 3}{99 \div 3} = \frac{4}{33}$

d) $0,454545... = \frac{45 \div 9}{99 \div 9} = \frac{5}{11}$

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e) $0,159159159... = \frac{159 \div 3}{999 \div 3} = \frac{53}{333}$

f) $0,564564564... = \frac{564 \div 3}{999 \div 3} = \frac{188}{333}$

g) $0,077777... = \frac{7}{90}$

h) $0,011111... = \frac{1}{90}$

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i) $0,006666... = \frac{6 \div 6}{900 \div 6} = \frac{1}{150}$

j) $0,0121212... = \frac{12 \div 6}{990 \div 6} = \frac{2}{165}$

k) $0,0232323... = \frac{23}{990}$

l) $1,33333... = 1 + 0,33... = 1 + \frac{3}{9} = \frac{12}{9} \div 3 = \frac{4}{3}$

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m) $12,565656... = 12 + 0,5656...$
 $12 + \frac{56}{99} = \frac{1244}{99}$

o) $1,353535... =$
 $1 + 0,35$
 $1 + \frac{35}{99} = \frac{134}{99}$

n) $1,111111... = 1 + 0,11...$
 $1 + \frac{1}{9}$
 $\frac{10}{9}$

↓ ↓ ↓
p) $0,0003333... =$
 $\frac{3 \div 3}{9000 \div 3} = \frac{1}{3000}$

$$\begin{array}{r} 12 \\ 99 \\ \hline 108 \\ 108 + \\ \hline 1188 \\ 1188 + \\ \hline 1244 \\ \hline \end{array}$$

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q) $2,001111... =$
 $2 + 0,0011...$
 $2 + \frac{1}{900} = \frac{1801}{900}$

r) $3,022222... =$
 $3 + 0,02$
 $3 + \frac{2}{90} = \frac{272}{90} \div 2 = \frac{136}{45}$

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GABARITO DAS QUESTÕES

- | | | | |
|-----------------------|----------------------|---------------------|---------------------|
| a) $\frac{2}{9}$ | b) $\frac{5}{9}$ | c) $\frac{4}{33}$ | d) $\frac{5}{11}$ |
| e) $\frac{53}{333}$ | f) $\frac{188}{333}$ | g) $\frac{7}{90}$ | h) $\frac{1}{90}$ |
| i) $\frac{1}{150}$ | j) $\frac{2}{165}$ | k) $\frac{23}{990}$ | l) $\frac{4}{3}$ |
| m) $\frac{748}{33}$ | n) $\frac{10}{9}$ | o) $\frac{134}{99}$ | p) $\frac{1}{3000}$ |
| q) $\frac{1801}{900}$ | r) $\frac{136}{45}$ | | |